

Aditi Kalle

Thermal-Fluids & Experimental Validation

Four years building, instrumenting and debugging high-heat-flux test hardware — boiling, capillary evaporation, CHF, and liquid-metal blanket materials.

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etadm@gmail.com · 217-200-3419 · [linkedin.com/in/aditikprofile](https://www.linkedin.com/in/aditikprofile)



magnetic mirror

Where I have worked

One coherent thread since 2021: heat transfer at surfaces that fail by drying out.

● 2021–22

UIUC

Energy & Multiphase Flow Lab

Pool boiling on micro-machined stainless; falling-film evaporation and CaCO_3 fouling.

Two co-authored papers — ASME JHMT and Desalination.

● 2023

UCSD

Thermal Energy Materials & Physics

Capillary-fed thin-film evaporation on AAO, glass fibre and hydrogel wicks.

CHF characterised to $\sim 600 \text{ W/cm}^2$ on GB-140.

● 2024

Zap Energy

Advanced Materials Group

Liquid-metal blanket alloys and porous refractory structures.

Eutectic found by STA/DSC; porous W permeability modelled and verified.

● 2025–

MarineSitu

DOE ORISE / WPTO Fellow

Virtual twin and LP/MILP scheduling for off-grid marine power.

Deployable schedules validated against ERA5 across five sites.

High heat flux and dryout

The failure mode I have spent four years measuring is the one that ends a plasma-facing surface.

What I actually did

- Ramped heat flux on capillary-fed wicks until the departure from linear q'' vs ΔT signalled incipient dryout.
- Framed the design limit as a pressure budget — Young–Laplace driving pressure against Darcy resistance.
- Built and ran the loops: degassing, flow control, high-density thermocouple arrays, pressure taps, Python DAQ.
- Root-caused the failures that were instrumentation rather than physics, and the ones that were physics.
- Separately, pool boiling on micro-grooved stainless — groove depth and pitch against nucleation and scaling.

~600

W/cm² CHF characterised

2

peer-reviewed papers

4

years of rig time

Materials and porous structures

Zap Energy, Advanced Materials Group — 2024



Liquid-metal blanket alloys

Research alloys prepared in an inert glovebox. Eutectic composition of a novel alloy family determined by STA with TGA and DSC.



Why the eutectic matters

Lowest melting point in the system and a single sharp freezing point — so the loop minimum temperature drops and nothing solidifies in a pipe.



Microstructure

SEM/EDX and optical microscopy to confirm phase morphology against the thermal analysis.



Porous tungsten

COMSOL porous-media modelling for porosity and permeability, with Archimedes density as the independent experimental check.

Compositions are proprietary — methods and reasoning described, specifics withheld.

Where this maps at Realta

A mirror puts the hardest heat-flux surfaces at the ends and in the heating systems, not at a divertor.

● End wall and expander

Concentrated charged-particle flux, spread only by how much magnetic expansion you can buy. The largest high-heat-flux surface in the machine.

Direct — CHF and dryout characterisation

● Gyrotron collectors and beam dumps

A 1 MW gyrotron dumps roughly its own output into the collector. CHF in subcooled flow boiling is the design limit.

Adjacent — same limit, different boiling regime

● Blanket

Neutron heating, tritium breeding, and — because the product is process heat — an outlet temperature that is a customer specification.

Direct — liquid-metal alloys, eutectics, porous refractory

● DEC electrodes

High heat flux on a surface held at high potential. Insulating breaks, coolant conductivity control, standoff.

New to me — and the problem I find most interesting

What I am proposing

Thermal-fluids test hardware for the Forge

- 01 Build the rigs**

Design, build and instrument the loops and calorimetry that qualify components before they go on a machine.
- 02 Get real data**

High-density thermocouple arrays, flow and pressure instrumentation, DAQ, and the discipline to tell an artefact from an effect.
- 03 Close the loop**

Correlate the thermal models against what the rigs actually measure, and say plainly what is verified versus inferred.

The Forge is being fitted out and Hammir construction starts this year. Available immediately.